

Question

The following five chemical processes exist:

1. Liquid N_2O_4 is a common non-aqueous solvent. Sodium azide can undergo a "strong base displacing a weak base" reaction in this solvent, similar to that of sodium ethoxide in water.
2. Determination of formaldehyde content in solution using iodine in sodium hydroxide alkaline conditions.
3. Silver nitrate reacts with a halogen element in potassium chloride solution to produce a mononuclear diamagnetic substance containing 48.39% silver.
4. Sodium reacts with molten sodium hydroxide to obtain a strong base commonly used in organic synthesis.
5. Industrially, PH_3 is prepared by the disproportionation reaction of white phosphorus in alkaline sodium hydroxide solution. In addition, another product is obtained, with a ratio to PH_3 of 3:1.

For the above processes, the sum of the simplest integer ratios of the coefficients of the balanced chemical equations are a, b, c, d, and e, respectively. Which of the following statements is correct for these five numbers?

- A. All other options are incorrect
- B. Odd numbers have 3.
- C. Even numbers have 2.
- D. All five numbers are greater than 5.
- E. Prime numbers have 3.

F. All five numbers are less than 10.

G. The sum of five numbers is 40.

H. The smallest number among five numbers is 3.

I. There exist 2 numbers among 5 numbers that are powers of 2.

Answer

Correct Answer: **I**

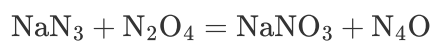
Detailed Explanation

For process 1, sodium ethoxide will react with water to produce sodium hydroxide and ethanol. Therefore, NaNO_3 and $[\text{NO}][\text{N}_3]$ will be formed, i.e., the balanced chemical equation is

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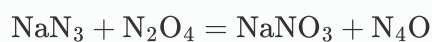
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Formation of NaNO_3 and $[\text{NO}][\text{N}_3]$



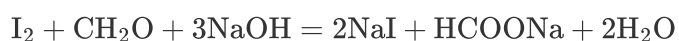
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The sum of coefficients is 4.

For process 2, according to the knowledge of elements, a redox reaction will occur, and formaldehyde is oxidized to formic acid. The balanced chemical equation is



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The sum of coefficients is 10.

For process 3, since the electronic configuration of Ag is $[\text{Kr}]4\text{d}^{10}5\text{s}^1$, the diamagnetic substance can only be Ag with a valence of +1 or +3. Considering that halogen element participates in the reaction, Ag should be oxidized to +3 valence, and the only halogen element with this oxidizing property is F. From the question "containing 48.39% silver", it is assumed that the product is K_xAgF_y , which should satisfy: $x + 3 = y$, $\frac{107.9}{107.9 + 39.1x + 19y} = 0.4839$, it can be judged that the product is KAgF_4 . Similarly, due to the strong oxidizing property of F_2 , Cl^- will be oxidized to Cl_2 . The balanced chemical equation is

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Halogen element participates in the reaction, Ag should be oxidized to +3 valence, only F_2

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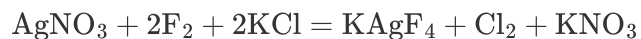
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The product is KAgF_4

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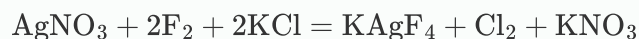
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Due to the strong oxidizing property of F_2 , Cl^- will be oxidized to Cl_2



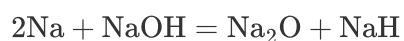
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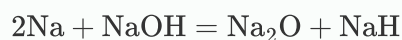
The sum of coefficients is 8.

For process 4, sodium can only act as a reducing agent in this reaction, so it can be inferred that the strong base is NaH, and the reaction equation is



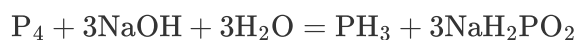
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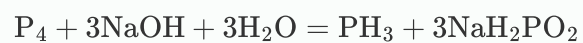
The sum of reaction coefficients is 5.

For process 5, according to the disproportionation reaction ratio, the oxidation state of phosphorus in the other product should be +1, and the existing form under alkaline conditions should be H_2PO_2^- . Therefore, the equation can be written as:



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The sum of coefficients is 11.

In summary, the five numbers are 4, 10, 8, 5, 11, option I is correct.